

Third-Person Epistemology: Information Isomorphism Epistemology and the Objective Foundation of Testability

—From "I Think" to "Anyone Running the Code": How an Ontological Fact, Rigorously
Calibrated on Truth-Known Systems, Proactively Verified on Truth-Unknown Systems,
Supported by Convergent Evidence Across Six Disciplines, and Independently Confirmed
Through Tool Replaceability, Legislates for Epistemology

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Abstract

This paper argues that Information Isomorphism Epistemology completes a paradigm shift in epistemology from the first person to the third person. Since Descartes, epistemology has been constructed in the first person—"I think," "I posit," "I see." When two "I"s reach conflicting conclusions, there is no public procedure to adjudicate. Information Isomorphism Epistemology is constructed in the third person: anyone, anywhere, downloading the publicly available data and running the same code, will obtain the same regime-switching structures. The argument proceeds from four mutually independent and mutually reinforcing pillars. First Pillar: The 2D Ising exact solution—the Chow F achieves a globally unique maximum at the precise phase transition temperature, with a deviation strictly zero in the analytic sense. Second Pillar: The 3D Ising double-peak fission—the algorithm, without any physical knowledge, proactively discovered a regime-switching structure previously unknown. Third Pillar: Cross-disciplinary emergence—the Factor Hierarchy Law has emerged repeatedly across six completely independent disciplines. Fourth Pillar: Tool replaceability—eight tools based on entirely different mathematical principles all converge near the confirmed breakpoints. These four pillars jointly point to a verified ontological fact: correct propositions leave detectable traces in statistical structures. From this fact, a criterion is derived: an epistemological system that makes factual claims about the empirical world must itself be testable. The legitimacy of this criterion does not derive from conceptual analysis but from the objective foundation jointly supported by these four pillars. Judged by this criterion, Descartes' "I think," Kant's "the thing-in-itself is unknowable," and Popper's falsifiability standard—all three have failed, in practice, to provide any public procedure satisfying the standard of traceability. Then, judged by the same criterion, Information Isomorphism Epistemology itself—each of its four components makes factual claims about the world, and each has passed the proposer's self-test. Information Isomorphism Epistemology is equally strict with itself, provides explicit falsification conditions, and bears the risk of being empirically overturned.

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1. A Question Epistemology Must Answer

Every epistemological system does the same thing: it tells others "what counts as reliable knowledge." It provides a standard, and then says: claims that satisfy this standard are reliable, and those that do not are not.

But there is a question that all epistemological systems must answer: **does the standard you provide satisfy itself?**

If the standard provided by an epistemological system does not satisfy itself—if it cannot be supported or refuted by any empirical evidence—then how can it tell others "what counts as reliable knowledge"? It teaches others how to distinguish truth from falsehood, while standing itself outside the adjudication of truth and falsehood. This is not epistemology; this is privilege.

The predicament of traditional epistemology lies in this: its claims can be simultaneously interpreted as asserting facts about the world and as proposing methodological conventions. Descartes' "I think" can be interpreted as asserting the existence of the self and its epistemological priority, or as recommending a method of doubt. Kant's "the thing-in-itself is unknowable" can be interpreted as asserting a universal boundary of human cognitive capacity, or as clarifying the rules of use for the concept of "experience." Popper's falsifiability standard can be interpreted as asserting a necessary condition for scientific knowledge, or as proposing a demarcation convention. When the authority of the standard needs to be established, they are interpreted as factual claims; when the legitimacy of the standard itself is questioned, they are interpreted as methodological conventions. This dual interpretability renders them neither testable nor trustworthy.

2. From the First Person to the Third Person

This paper is not proposing a new viewpoint within the framework of traditional epistemology. It is a thorough transformation of the subject position of epistemology.

Descartes founded epistemology on the first person—"I think." For nearly four centuries thereafter, epistemology has been a variation of the first person: Kant's first-person reason reflecting on its own conditions, Popper's first-person reason setting demarcation standards. Traditional epistemology has always relied on a special "I" to lay the foundation of knowledge—an "I" possessing a privileged access to truth, whether through introspection, transcendental reflection, or conceptual intuition. When two "I"s reach conflicting conclusions, there is no public procedure to adjudicate who is right and who is wrong.

Information Isomorphism Epistemology switches epistemology from the first person to the third person. It is not "I see," "I think," "I posit," but "a public procedure that anyone can run," "the same result that anyone running the code will obtain." The subject of epistemology is no longer the philosopher with privileged access, but any person. The standard no longer requires anyone to "see" into the proposer's mind—it only requires anyone to run the code.

This is not evading the core questions of traditional epistemology, but answering them with new methods. Traditional epistemology asks "how is knowledge possible," but its answers—transcendental apperception, forms of intuition, categories—cannot be tested. Asking a question that cannot be tested is legitimate, but providing an answer that cannot be tested is not. Information Isomorphism Epistemology precisely answers "how knowledge is possible"—not through speculative reasoning, but through a verified ontological fact: correct propositions leave detectable traces in statistical structures, and statistical structures are publicly detectable. Third-person epistemology is not diminishing the task of epistemology—it is giving that task a testable answer. The complete argument for this paradigm shift and its philosophical consequences for Kantian agnosticism has been systematically developed in Tang (2026ad). This paper completes, on that foundation, the ultimate transition of epistemology from the first person to the third person—not arguing that "the thing-in-itself is no longer necessary," but arguing that "epistemology must itself be testable."

The argument of this paper blends two different modes of argumentation—philosophical argument and scientific argument. The former relies on the rigor of logical deduction, the latter on the replicability of facts. These two standards are different but equally strict. The ultimate conclusion of this paper satisfies both: the logical derivation is valid, and the core facts are replicable.

3. The Establishment of the Criterion: From Four Pillars to the Requirement of Epistemology

Information Isomorphism Epistemology consists of four components: the Law of Information Isomorphism, the Factor Hierarchy Law, the Principle of Testability, and the Norms of Testability (Tang, 2026g, 2026r, 2026z). This section argues for the empirical foundation of this entire system's effectiveness—not conceptual analysis, not philosophical presupposition, but empirical evidence from four independent dimensions.

3.1 The Shared Orientation of the Four Pillars

Information Isomorphism Epistemology grew out of repeated deployment across six disciplines. After its completion, one can look back and ask: what makes this system effective? Evidence from four independent dimensions jointly points to the same answer.

First Pillar: The 2D Ising Exact Solution—on a truth-known system, information isomorphism is precise. On the Onsager exact solution of the 2D Ising model (Onsager, 1944), the Chow F, without any physical knowledge, achieves a globally unique maximum ($F = 118,074$) at the precise phase transition critical temperature $T_c = 2.269185$, with a deviation strictly zero in the analytic sense (Tang, 2026k). The intercept/slope ratio is strictly monotonic with distance from T_c ($\rho = 1.0000$), and is strictly negatively correlated with both fluctuation intensity and mean-field description error ($\rho = -1.0000$) (Tang, 2026x, Theorem A). These two information-theoretic dualities—the Chow F–order parameter duality and the intercept/slope ratio–critical distance duality—respectively establish that "statistical breakpoints precisely correspond to regime boundary positions" and that "statistical signatures precisely correspond to physical critical behavior" (Tang, 2026ac). This fact establishes the upper bound of precision for information isomorphism: on a truth-known system, the correspondence between statistical structures and regime structures is precise. In this pillar, the correctness of the T_c position given by the Onsager exact solution is entirely independent of the Chow F—physicists derived T_c algebraically in 1944, and the Chow F was applied later for verification. There is no circularity here.

Second Pillar: 3D Ising Double-Peak Fission—on a truth-unknown system, information isomorphism can be proactively discovered. On the 3D Ising model—a problem strictly proven to have no analytic solution and to be NP-complete—the algorithm, without any physical knowledge, proactively discovered a double-peak fission structure never previously revealed by traditional finite-size scaling analysis: the Bai-Perron double-breakpoint blind test automatically locked $T_1 = 4.39$ and $T_2 = 4.49$, precisely aligning with the switch of spontaneous magnetization M_{sp} and the switch of the Binder cumulant U_4 (Tang, 2026l). This discovery was later independently replicated on the 2D Potts $q = 4$ model (Tang, 2026m). The confirmation of double-peak fission as a real physical structure depends in part on the calibration transmission from 2D Ising and on post-hoc independent verification: the algorithm was calibrated on 2D Ising, hence its discovery on 3D Ising has a calibration basis; and the physical quantities used for post-hoc verification (M_{sp} , U_4) are independently computed through Monte Carlo simulation, independent of the Bai-Perron structural breakpoint testing tool. The First Pillar provides a completely independent calibration benchmark; the Second Pillar demonstrates, on that basis, proactive discovery capability. Together, the two pillars constitute a complete argumentative chain from "precision on truth-known" to "discovery power on truth-unknown."

Third Pillar: Cross-Disciplinary Emergence—information isomorphism is universal. The Factor Hierarchy Law—a meta-law of meta-laws that subsumes derived meta-laws such as Noether's theorem, the principle of least action, and Shannon's information theory (Tang, 2026r)—first emerged in financial data: the direct incremental R^2 of the China-US interest rate spread was close to zero, but the interaction incremental R^2 was 1.78 times the direct incremental R^2 (Tang, 2026e). It has since been repeatedly verified across five completely independent disciplines. The evidential strength of these verifications is not uniform. Post-hoc

verifications prove the precision of information isomorphism: in particle physics, the algorithm, without any knowledge of the physical meaning of charm quarks, automatically locked onto the positions of charmed baryons and bottomonium—the charm-region breakpoint deviated from the charmed baryon standard value by only 0.00004 GeV, and the bottom-region breakpoint deviated from the bottomonium standard value by only 0.0001 GeV (Tang, 2026n). Proactive discoveries prove the discovery power of information isomorphism: in astronomy, stellar spectral type triggers a regime switch in the transit depth equation at 4762K—the Tang Break, with five independent physical dimensions all structurally ruptured (Tang, 2026h–j); in chemistry, the operating system provided purely deductive numerical predictions for yet-unsynthesized elements (Tang, 2026t); in nuclear physics, N and Z were confirmed as semi-continuous regime factors, $\eta > 10^6$, and $N = 14$ was cross-validated by three independent tools as a statistically significant structural rupture—a position never previously marked by traditional nuclear physics (Tang, 2026v). Post-hoc verification establishes precision; proactive discovery establishes discovery power. Six independent disciplines, six independent datasets, the same testing procedures, the same structural emergence—this cannot be explained as an artifact produced by the method itself, because fake data audits output blanks.

Fourth Pillar: Tool Replaceability—information isomorphism does not depend on specific tools. In nuclear physics, eight tools based on entirely different mathematical principles—the Chow test, Bai-Perron, CUSUM, MOSUM, PELT, Bayesian change point detection, HMM, and kernel change point detection—all converge near the confirmed breakpoints (Tang, 2026w). Activation operator stratification is completely isomorphic on the N -axis and the Z -axis (Tang, 2026aa). The legitimacy of this pillar is rooted in the cross-validation rule of the Norms of Testability—the structural breakpoint dimension requires independent testing by at least two tools based on different classifications of mathematical principles (Tang, 2026z). If the choice of statistical quantities is theory-laden—different statistical tools may produce different statistical structures—then tool convergence is a direct response to this challenge: tools based on different mathematical principles all converge near the confirmed breakpoints, which means that the signal of regime switching is not an artifact of a specific tool.

These four pillars are mutually independent and mutually reinforcing. The First Pillar answers "how precise can it be," the Second Pillar answers "can it discover in unknown domains," the Third Pillar answers "is it universal or local," and the Fourth Pillar answers "does it depend on specific tools." The shared orientation of the four pillars is: **correct propositions leave detectable traces in statistical structures.**

3.2 From Fact to Principle: Calibration Benchmark and Operational Definition

From the shared orientation of the four pillars, an operational definition of "testable" can be established.

The 2D Ising exact solution provides a truth-known calibration benchmark. On this benchmark, the performance of "correct" physical facts and "incorrect" physical facts on the Chow F exhibits detectable

differences—the Chow F achieves a globally unique maximum at T_c and drops sharply when deviating from T_c . The 3D Ising double-peak fission further proves that such differences can be proactively discovered. Cross-disciplinary emergence proves that such differences are not limited to specific physical systems. Tool replaceability proves that such differences do not depend on specific statistical tools.

From these facts, an operational definition of "testable" can be extracted: there exists a public procedure satisfying the standard of traceability, whose performance on a truth-known calibration benchmark is significantly better than random levels. Traceability is the defining characteristic of a public procedure: when two independent subjects executing the same procedure obtain different results, there exists a traceback path—checking the code, checking the data, checking the random seed—step by step until the point of divergence is found.

This operational definition bears the risk of being empirically overturned. Its reasonableness lies in this: if one does not accept it, one must accept that "there exist unknowable truths"—and unknowable truths, in epistemological practice, cannot be used to calibrate procedures.

The status of 2D Ising as a calibration benchmark, being "truth-known," depends on the collective acceptance by the physics community of Onsager's algebraic derivation. This dependence differs fundamentally from traditional epistemology's dependence on authority: Onsager's derivation is public, and anyone with sufficient mathematical training can independently verify it. More importantly, if the verification of Onsager's derivation fails, there exists a traceback path—checking whether each step of the algebraic transformation is correct. Descartes' introspection, Kant's transcendental argument, and Popper's conceptual analysis are in principle impossible to verify—because the content of introspection cannot be publicly traced back, the premises of transcendental argument cannot be publicly stipulated, and the intuitions of conceptual analysis cannot be publicly calibrated.

Currently, the 2D Ising exact solution is the only truth-known calibration benchmark that is completely independent of the testing procedures of this paper. Any methodological system must start its calibration from some benchmark. The 2D Ising exact solution is suitable as a starting point because it is the cleanest truth-known system currently available: the algebraic derivation is independent of statistical tools, the solution is exact, and anyone with sufficient mathematical training can independently verify it. If calibration is completed on other truth-known systems in the future, calibration benchmarks will move from singularity to plurality, and the objective foundation of the criterion will be further strengthened.

The criterion of this paper requires the claimant to provide or point to a truth-known calibration benchmark, such that the testing procedure can be calibrated on that benchmark. The only explicit standard is traceability—the truth of the calibration benchmark must be independently verifiable through a publicly traceable procedure. But traceability is a continuous spectrum: the traceability of mathematical derivation is strictly exact, while the traceability of empirical regularities involves theoretical assumptions and data quality. At what degree of

traceability a benchmark can be recognized as "truth-known" is formalization work not yet completed in this paper. Any methodology, at its inception, must start from the cleanest benchmark. As more calibration benchmarks are incorporated, the very standard for recognizing calibration benchmarks will itself be calibrated—a recursive process of self-improvement.

This paper does not claim that 2D Ising is the only benchmark that can calibrate the operational definition of "testable," nor does it claim that the current operational definition is the final version. Anyone in the future may propose new calibration benchmarks or new operational definitions. This is precisely the superiority of public procedures over conceptual analysis: competition between conceptual analyses has no adjudication procedure; competition between operational definitions has one—namely, the calibration benchmark.

3.3 The Derivation and Structure of the Criterion

From the above operational definition, a criterion can be derived: an epistemological system that makes factual claims about the empirical world must itself be testable.

The legitimacy of this criterion does not derive from conceptual analysis. It is a logical derivation from a verified ontological fact jointly pointed to by the four pillars. Testing is not a standard invented by humans—it is a requirement of the world itself. This turn from fact to norm is rooted in the relationship between the principle of information isomorphism and testability established in Tang (2026r): the principle of information isomorphism provides the epistemological foundation for testability—precisely because there exists a precise information correspondence between statistical structures and regime structures, regime-dependence testing can capture genuine signals of regime switching.

The derivation from "correct propositions leave detectable traces in statistical structures" (descriptive) to "an epistemological system must itself be testable" (normative) depends on an instrumental rational presupposition: if one wants to obtain reliable knowledge, one should use methods that can distinguish the correct from the incorrect. This presupposition is the premise of any epistemological enterprise. If a person does not want reliable knowledge, the criterion of this paper has no binding force on them. But if a person claims to be providing standards for reliable knowledge, then they have already committed to pursuing reliable knowledge—and pursuing reliable knowledge requires using methods that can distinguish the correct from the incorrect. What the Law of Information Isomorphism reveals is precisely the basis for such distinction: correct propositions leave detectable traces in statistical structures. Therefore, an epistemological system that makes factual claims about the empirical world—if it truly pursues reliable knowledge—must itself be testable.

The criterion has an "if-then" conditional structure: if the Law of Information Isomorphism holds, then an epistemological system must itself be testable. The criterion itself is a normative conclusion, and its being tested is accomplished indirectly through the survival or demise of its premise—the Law of Information Isomorphism. This "being tested" differs from the direct empirical testing of descriptive propositions—it is the indirect

empirical constraint that a normative proposition obtains through the objective foundation of its premise. This paper does not claim that the criterion derived from the four pillars is the only possible epistemological criterion. It claims only: among all currently known epistemological criteria, this criterion is the only one derived from a verified ontological fact, and therefore it is currently the best-founded criterion. If other criteria can be derived from more solid objective foundations in the future, the criterion of this paper can be replaced—this is precisely the embodiment of the self-correctability of this paper's criterion.

The criterion of this paper uses "testing" in its most general sense: the reliability of a claim does not depend on the authority of the claimant, but on a public procedure that any independent third party can execute and that satisfies the standard of traceability. The failure of traditional epistemology lies not in its not having used scientific methods to test its claims—but in its not having provided any public procedure satisfying the standard of traceability. Introspective self-evidence does not satisfy traceability—when two people introspect differently, there is no traceback path. The "being seen" of transcendental argument does not satisfy traceability—when Kant and Heidegger "see" different transcendental conditions, there is no traceback path. The "intuition" of conceptual analysis does not satisfy traceability—when two philosophers have different analyses of the same concept, there is no traceback path.

Why must this ontological fact be elevated to an epistemological criterion? It could simply be accepted as a methodological discovery. But if it is accepted only as a methodological discovery, traditional epistemology can continue to evade adjudication under the protection of dual interpretability. The function of the criterion is not to increase the truth value of this fact—the truth value of the fact does not depend on the criterion—but to make the epistemological consequences of this fact impossible to ignore.

3.4 Self-Examination of the Criterion

The criterion of this paper imposes a threshold on claims—the claimant must provide or point to a truth-known calibration benchmark, such that the testing procedure can be calibrated on that benchmark. If a claim has no truth-known calibration benchmark—such as "the thing-in-itself is unknowable"—then it cannot even enter the calibration procedure. This is not a failure of the criterion, but a fundamental difference between the claims of traditional epistemology and scientific claims: the former are designed to require no calibration benchmark, while the latter must be calibrated.

Does the core criticism of traditional epistemology made by this paper—that dual interpretability renders it neither testable nor trustworthy—also apply to the criterion of this paper? The criterion of this paper can be interpreted as a factual claim (its premise being the Law of Information Isomorphism), or as a methodological convention (a pragmatic recommendation based on instrumental rational presupposition). If interpreted as a factual claim, its condition of being overturned is the falsification of the Law of Information Isomorphism. If interpreted as a methodological convention, its condition of being replaced is the proposal of a better-founded

criterion by someone else. Under either interpretation, the criterion has a path by which it can be overturned or replaced. The core claims of traditional epistemology provide no conditions under which they could be overturned. This is not a problem of "dual interpretability"—that a claim can be interpreted in multiple ways is not itself a problem. The problem is whether there exists an irremovable immunity mechanism when legitimacy is questioned. Traditional epistemology has this mechanism; Information Isomorphism Epistemology does not.

Is the Principle of Testability itself an untestable metaphysical presupposition? It is not. It is not an externally imposed metaphysical presupposition, but an operational necessity that emerged internally from the discovery of the Factor Hierarchy Law—when Tang (2026e) discovered that "the function of some factors is not to directly explain the dependent variable, but to regulate the exposure coefficients of other factors," a methodological question naturally arose: how to distinguish these two types of factors? The Principle of Testability is the answer to this question. It is not "I decree that science must do this," but "if I do not do this, even I myself cannot distinguish true discoveries from false positives" (Tang, 2026g).

4. The Adjudication of Traditional Epistemology

Epistemology is not autobiography. Descartes reported his inner experience while doubting, Kant reported reason's critique of itself, Popper reported his intuition about scientific demarcation. These reports may all be honest—but the reporter cannot demand that others treat his inner experience as a universal standard. Epistemology—if it makes claims about facts in the world—should be a publicly traceable procedure.

Now, let us judge traditional epistemology by the criterion of this paper. The standard of judgment is one alone: does this system provide a public procedure satisfying the standard of traceability, such that an independent third party can adjudicate whether its core claims hold?

Descartes. Descartes' "I think" can be interpreted in two ways. Interpreted as a factual claim: it asserts the existence of the self and its epistemological priority. Interpreted as conceptual necessity: it reveals that the concept of "doubt" itself contains the logical necessity of "existence." Descartes' strongest defense is neither of these interpretations—but rather: "I think" is not a claim; it is an execution. The act of "I am doubting" itself, when you execute it, your existence is given in this act. The validity of this defense ends where? At the traceback path when execution fails. If two people execute doubt and reach different conclusions, Descartes would say "the first person executed correctly, the second did not truly execute"—but this adjudication standard itself is private. The traceback path of Information Isomorphism Epistemology in the same situation is public: check the code, check the data, check the random seed. The difference lies not in the form of "inviting execution," but in whether there is a traceback path when execution fails.

Kant. Kant's "the thing-in-itself is unknowable" is the conclusion of a transcendental argument. Kant's strongest defense is: transcendental argument traces the necessary conditions of the possibility of experience.

This traceback is public—if one rejects "the forms of intuition are space and time," one must explain why geometry is universally and necessarily valid for the physical world. Where does the publicity of this defense end? At the premise that "geometry is universally and necessarily valid for the physical world." If you accept this premise, you can examine Kant's reasoning. If you do not—if you believe, for instance, that physical space is non-Euclidean—then Kant's argument has no binding force on you. The premises of transcendental argument are claimed to be necessary structures "discovered" in rational reflection. When a person does not "discover" this structure in reflection, the defender of transcendental argument can only say that the person has not reflected deeply enough—this is an adjudication with no traceback path. The premises of public procedures are "stipulated" axioms—you do not need to "discover" Euclid's postulates in reason; you only need to know what they are. If you do not accept them, you can switch to a different set of axioms. Consensus is discovered; stipulation is created. The traceability of transcendental argument is limited to those who share the same philosophical intuition—they happen to have "discovered" the same premises. The traceability of public procedures crosses the boundary of intuition.

Popper. Popper's falsifiability standard claims that falsifiability demarcates the boundary of science. If this is only a methodological convention, it cannot claim to be "correct"—only "useful." But Popper did claim that falsifiability corresponds to the actual boundary of scientific practice—this is already a factual claim about the world. Yet he never provided a public procedure satisfying the standard of traceability to test this claim—you cannot test falsifiability itself with falsifiability. Popper's falsifiability guards the entrance of science, but he never provided a standard to guard the exit. Testability is precisely the standard for the exit—and Information Isomorphism Epistemology satisfies this standard (Tang, 2026g).

This paper has adjudicated three of the most representative systems of traditional epistemology. They represent, respectively, rationalism (Descartes), transcendental idealism (Kant), and critical rationalism (Popper). Naturalized epistemology and experimental philosophy, discussed in Note 1, have likewise failed to provide public procedures satisfying the standard of traceability to test their core claims. Whatever traditional epistemologists call their claims—factual claims, transcendental necessities, conceptual truths, demarcation conventions—they share, in practice, a common feature: when two systems that equally claim necessity or conceptual truth reach different conclusions, there is no traceback path to adjudicate which is right and which is wrong. Their reliability ultimately depends on the authority of the arguer.

5. The Self-Adjudication of Information Isomorphism Epistemology

Now, let us judge Information Isomorphism Epistemology itself by the same criterion. Traditional epistemology judges others and exempts itself. Information Isomorphism Epistemology judges others, and also judges itself—and is stricter with the latter. Information Isomorphism Epistemology consists of four

components. Each makes factual claims about the world. Each must submit to testing. The testing results reported in this section have been completed by the proposer; all testing procedures and data have been made public, awaiting independent third-party replication.

The Factor Hierarchy Law. The Factor Hierarchy Law claims: any complex system is organized by a hierarchical structure of regime factors and execution factors. Sub-claim (1): The function of regime factors is embodied through interaction effects. Testing standard: whether the ratio of interaction effects to direct effects systematically deviates from zero. Testing results: In the initial deployment in finance, the interaction incremental R^2 was 1.78 times the direct incremental R^2 (Tang, 2026e); in nuclear physics, $\eta > 10^6$ (Tang, 2026v); repeatedly verified across five completely independent disciplines. Fake data audits output blanks. Proposer self-test: passed. Sub-claim (2): When a regime factor crosses a critical value, a regime switch occurs. Testing standard: on a truth-known system, whether the interaction effect peaks at the regime boundary. Testing results: The Chow F achieves a globally unique maximum at the precise T_c , with deviation strictly zero in the analytic sense. On a truth-unknown system, the algorithm proactively discovered a previously unknown regime-switching structure—the 3D Ising double-peak fission (Tang, 2026l). Proposer self-test: passed. The Factor Hierarchy Law encompasses three forms—pure regime factors (interaction effects significant but direct R^2 low), dual-identity factors (interaction effects significant and direct R^2 high), and pure execution factors (direct R^2 high but interaction effects not significant) (Tang, 2026g). All three forms have been verified across the six disciplines.

The Norms of Testability and the Principle of Testability. The Norms of Testability claim: the systematic integration of six functional dimensions can detect regime switches. Testing standard: whether fake data audits output blanks; whether permutation tests confirm that signals are not random noise. Testing results: Fake data audits output blanks; permutation tests confirm that signals are not random noise. Given the same input data and random seed, the determinacy of the code guarantees that anyone running it will obtain the same output. Proposer self-test: passed. The Principle of Testability claims: any claim must pass testing to be reliable. Testing standard: whether its scope of applicability and formulation can be empirically revised. Testing results: If a system is discovered in which regime switches can be reliably confirmed without testing, the scope of applicability of this principle would be revised. Proposer self-test: passed.

The Law of Information Isomorphism. The Law of Information Isomorphism claims: statistical structures carry information about regime structures. This is a dual meta-law: it is simultaneously an ontological law and an epistemological meta-law—it provides the epistemological foundation for the Factor Hierarchy Law, and the two are complementary (Tang, 2026ac). Testing standard: on a truth-known system, whether the position of the regime boundary is precisely encoded as the position of the statistical breakpoint. Testing results: On the 2D Ising exact solution, the standard is precisely satisfied—deviation strictly zero. On the 3D Ising

model, the algorithm proactively discovered a previously unknown double-peak fission structure. Across six disciplines, convergent evidence supports this law. Proposer self-test: passed.

Self-Referentiality Test. The Factor Hierarchy Law obeys itself: its "data"—the verification records across six disciplines—are produced by automated procedures, satisfying the condition of objective data; its "regime factor" is "whether the data are objective"; its "execution factors" are the testing procedures themselves (Tang, 2026r). The objectivity condition of Information Isomorphism Epistemology—"data must be objective"—can itself be tested: fake data audits output blanks, confirming that the data satisfy the objectivity condition. Proposer self-test: passed.

All four components of Information Isomorphism Epistemology make factual claims about the world. Each has passed the proposer's self-test. What traditional epistemology fails to pass is the most general requirement of testing—"is there any public procedure satisfying the standard of traceability that can adjudicate?" What Information Isomorphism Epistemology passes is the strictest testing—the six functional dimensions of the Norms of Testability. The manner in which Information Isomorphism Epistemology passes testing is by using its own testing procedures. This does not constitute a logical circularity, because the Norms of Testability themselves have already passed strict negative controls: fake data audits output blanks, permutation tests confirm that signals are not random noise. A rigorously calibrated testing procedure, used to test the claims of the system to which it belongs, is not circularity—it is self-consistency.

The core integrative claim of Information Isomorphism Epistemology—"these four components together constitute a complete third-person epistemology that is more reliable than traditional epistemology"—is tested differently from the individual testing of the four components. The testing of the four components is statistical-structure testing, which can be rigorously calibrated on truth-known calibration benchmarks. The testing of the integrative claim is historical testing: can it consistently produce reliable new discoveries in future scientific practice, can it be confirmed through independent third-party replication, can it demonstrate stronger discovery power in comparison with other epistemological proposals. This kind of testing has no single truth-known calibration benchmark—its calibration benchmark is the historical process of scientific practice itself.

The skill threshold is crossable—a person can learn programming, can learn statistics; this knowledge is available in any standard curriculum. Descartes' introspection threshold, by contrast, is in principle uncrossable—if a person does not "see" the certainty of the "I think" in introspection, no one can teach them how to "see" it. The difference between Information Isomorphism Epistemology and traditional epistemology in adjudication lies not only in that the former passes while the latter does not—but more fundamentally in that the former is willing to submit to adjudication, while the latter has never placed itself in the position of being adjudicated. Information Isomorphism Epistemology adjudicates itself, and uses stricter standards than it uses to adjudicate others. This asymmetry is the substance of the argument. A system that is strict with others and stricter with itself is true epistemology.

6. Conclusion

Epistemology since Descartes has been first-person—when two "I"s conflict, there is no public procedure to adjudicate. Information Isomorphism Epistemology is third-person—when the results of two "anyones" conflict, there exists a traceback path: check the code, check the data, check the random seed, step by step until the point of divergence is found. This is the fundamental difference between third-person epistemology and first-person epistemology.

The Law of Information Isomorphism reveals that correct propositions leave detectable traces in statistical structures. This fact has been jointly verified on the four pillars—not dependent on anyone's authority. Traditional epistemology uses conceptual analysis to establish standards; these standards are artificially defined, and their legitimacy depends on the authority of the definer. Information Isomorphism Epistemology uses objective laws to establish standards. This transition means that epistemology changes from a discipline dependent on authority to a discipline dependent on objective laws. The greatest debates in the history of philosophy—rationalism versus empiricism, idealism versus realism, foundationalism versus coherentism—none has been adjudicated by a public procedure. Each was won by one authority persuading enough people, and then another authority emerged to restart the debate. Third-person epistemology, for the first time, makes epistemology no longer an arena of authority.

In the nearly four centuries since Descartes, the representative systems examined in this paper—representing rationalism (Descartes), transcendental idealism (Kant), and critical rationalism (Popper)—along with naturalized epistemology and experimental philosophy discussed in Note 1, have all failed to provide public procedures satisfying the standard of traceability to test their core claims. Their reliability ultimately depends on the authority of the arguer. Information Isomorphism Epistemology is equally strict with itself, provides explicit falsification conditions, and bears the risk of being empirically overturned.

The core claim of this paper—that Information Isomorphism Epistemology is the first epistemological system that is itself testable—can itself be tested. The traceback path is public: examine the operational definition of "itself testable" in this paper, examine whether traditional epistemology satisfies that definition, examine whether Information Isomorphism Epistemology satisfies that definition. Each step can be traced and adjudicated by any independent third party.

The argumentative form of this paper is itself an instance of its conclusion. The testing method for philosophical argument is logical traceback—examining whether the premises hold and whether the reasoning steps are valid. The philosophical argumentation portion of this paper satisfies the same standard of traceability: the premises are public, the reasoning steps are public, the traceback path exists. If a reader has objections to the derivation of the criterion, they can trace back step by step: check whether the Chow F peak in the First Pillar is indeed at T_c , check whether the Bai-Perron double breakpoints in the Second Pillar indeed align with the switches of M_{sp} and U_4 , check whether the fake data audit indeed outputs blanks. Each step of traceback can

be executed by any independent third party. This paper not only argues that "correct propositions leave detectable traces in statistical structures"—it itself demonstrates, in the dimension of philosophical argument, how "traceability" applies to logical derivation.

Information Isomorphism Epistemology has currently passed the proposer's self-test; all testing procedures and data have been made public, and it is awaiting independent third-party testing. Before independent testing is completed, the status of its claim to be "itself testable" is "self-test passed, awaiting external verification." This status itself is already superior to all traditional epistemology—because traditional epistemology does not even have a self-test procedure. The proposer explicitly invites any independent researcher to replicate. If replication fails, the core claim of this paper is falsified. If replication succeeds, the core claim of this paper is independently verified. Whichever the outcome, this paper has accomplished what traditional epistemology has never done: subjecting the claims of epistemology to testing by public procedures independent of the proposer.

As a philosophical argument, the validity of this paper's reasoning can be tested right now by any independent third party. The truth value of this paper's premises will be adjudicated by the historical process of scientific practice, but the validity of its derivations does not depend on the truth value of those premises—it is an "if-then" structure: if the premises hold, then the conclusion holds. If independent replication confirms the premises, the conclusion of this paper changes from "if-true" to "true." If independent replication falsifies the premises, the conclusion of this paper is not thereby proven invalid—it has merely lost a true premise. As an "if-then" derivation, this paper is complete as a philosophical argument right now.

Note 1

Naturalized epistemology (Quine, 1969) and experimental philosophy, although borrowing empirical methods, have their own core normative claims—"epistemology should be naturalized," "philosophical methodology should be based on experimental evidence"—still not independently tested by any public procedure satisfying the standard of traceability. The uniqueness of Information Isomorphism Epistemology lies in this: its core criterion is derived from a verified ontological fact, rather than proposed from conceptual analysis or methodological preference. Therefore, naturalized epistemology does not constitute a counterexample to the thesis of this paper.

Data Availability Statement

This paper is a theoretical overview and philosophical argumentation paper. All empirical data come from the series of preprints by Tang (2026a–2026ad). The rigorous proof of the Law of Information Isomorphism is detailed in Tang (2026k) and Tang (2026x) Theorem A. The discovery process of the Factor Hierarchy Law is detailed in Tang (2026e) and Tang (2026g). The discovery of the 3D Ising double-peak fission is detailed in Tang (2026l). The theorem of tool dependence is detailed in Tang (2026m) and Tang (2026w). The complete technical specification of the Norms of Testability is detailed in Tang (2026z). All data are completely public, and any researcher can independently replicate all results.

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